Renewable Energy Generation Forecasting and Decarbonization

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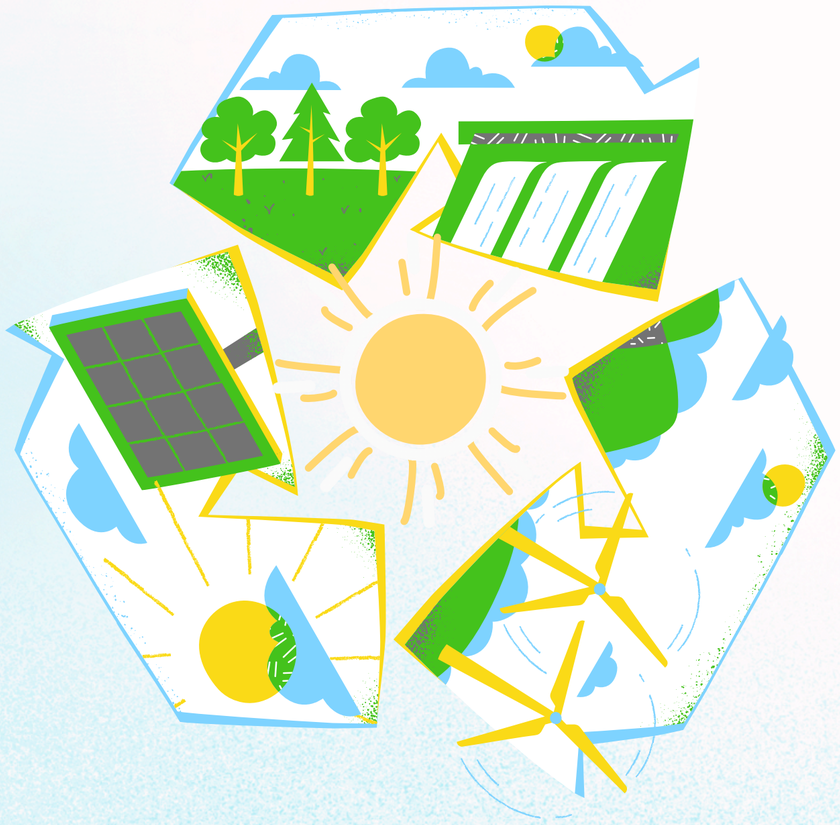


GAP ANALYSIS FOR

NEW YORK CONTROL AREA (NYCA)

Machine Learning for Environmental Engineering

MOTIVATION & BACKGROUND

MOTIVATION

Renewable generation sources, primarily wind and solar have highly variable output dependent on meteorological conditions

Urgent need to forecast the generation from renewable sources from fossil fuels to renewables sources

CURRENT STATUS

Renewable generation currently meets demand for 0% of hours, creating a persistent supply deficit.

Average hourly deficit is ~13,000 MWh.

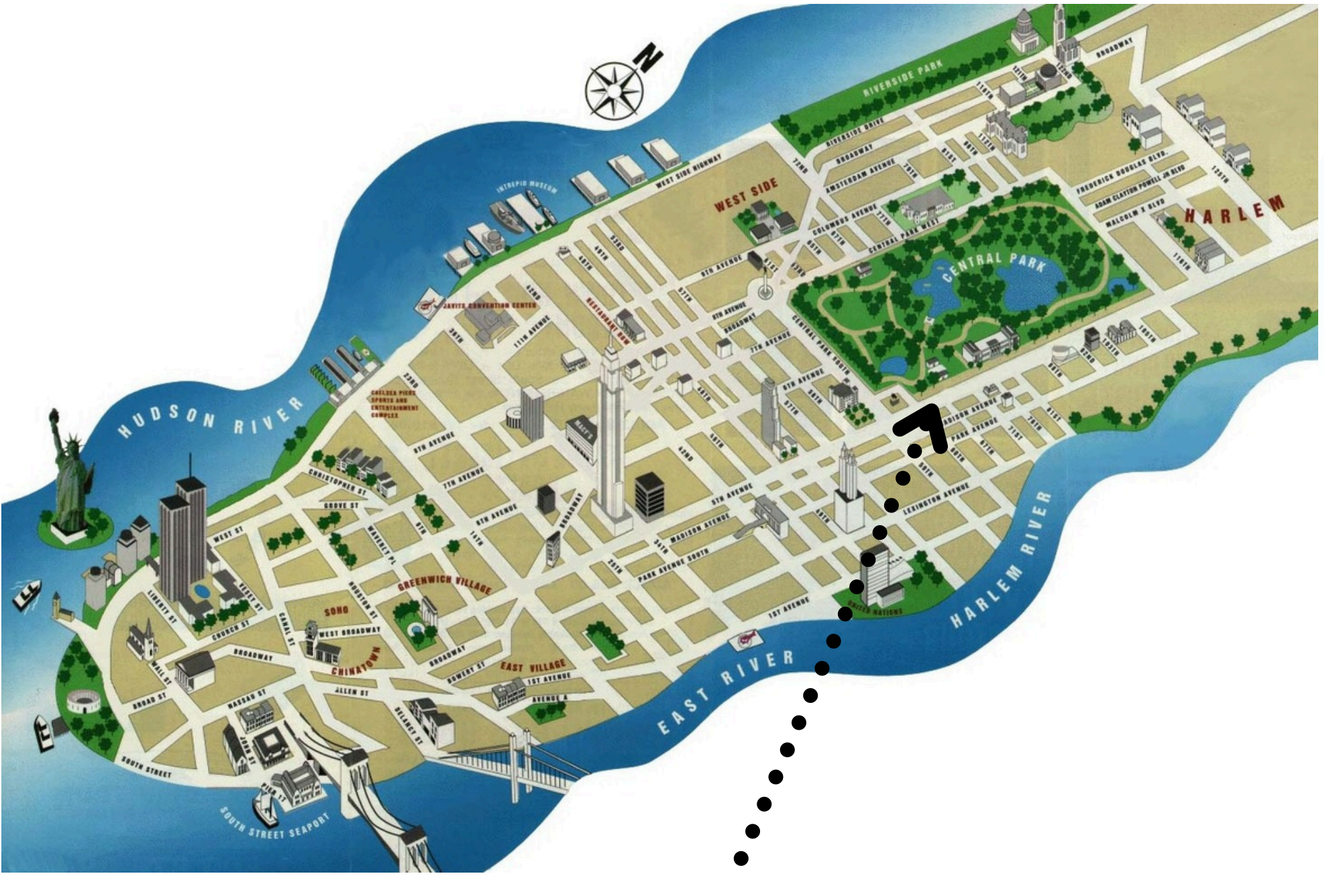
CHALLENGES

<u>Operational Challenge:</u>
Grid instability and reliability risk due to intermittent renewable generation.

<u>Environmental Challenge:</u>
Reliance on fossil fuels to fill the gap results in significant CO_2 emissions, hindering decarbonization

NYCA faces unprecedented challenges in balancing growing electricity demand with the intermittent nature of renewable energy generation.

PROJECT OBJECTIVE



Primary Weather Station (Jan 2020 - Nov 2025)

AIM

- Develop Advanced Forecasting Models
- Quantify Supply-Demand Gaps
- Capacity Expansion Requirement

METHODOLOGY MAPPING

A complete journey from raw data to deployment

STEP 01

Defining the Problem

STEP 02

Data sources and collection

Weather dataset
Generation & Load Dataset
Data quality assessment

STEP 03

Preprocessing and Integration

Data cleaning
Temporal Alignment
Feature engineering

STEP 04

Exploratory Data analysis

Getting key correlation findings

STEP 05

Train test Split strategy

Employment of train test split data

STEP 06

Results

Launch the model and continuously observe how it behaves in real-world conditions.

Data Sources and Collection

This analysis integrates two primary datasets spanning January 2020 to November 2025:

Weather Dataset (Central Park)

Generation and Load Dataset (NYISO)

- Total records: 51,555 hourly observations
- Date range: 2020-01-01 to 2025-11-19
- Weather data shape: (51,562 records, 13 variables)
- Load data shape: (51,606 records, 95 variables)
- Missing values: 52,719 instances (minimal, <0.2% of dataset)
- Data handling: Forward-fill interpolation applied to missing values
- Duplicates identified and removed: 6 duplicate records

Target Variable Definition

Total renewable generation calculated as:

Renewable Generation = Hydro + Wind + Solar + Geothermal + Other Renewable

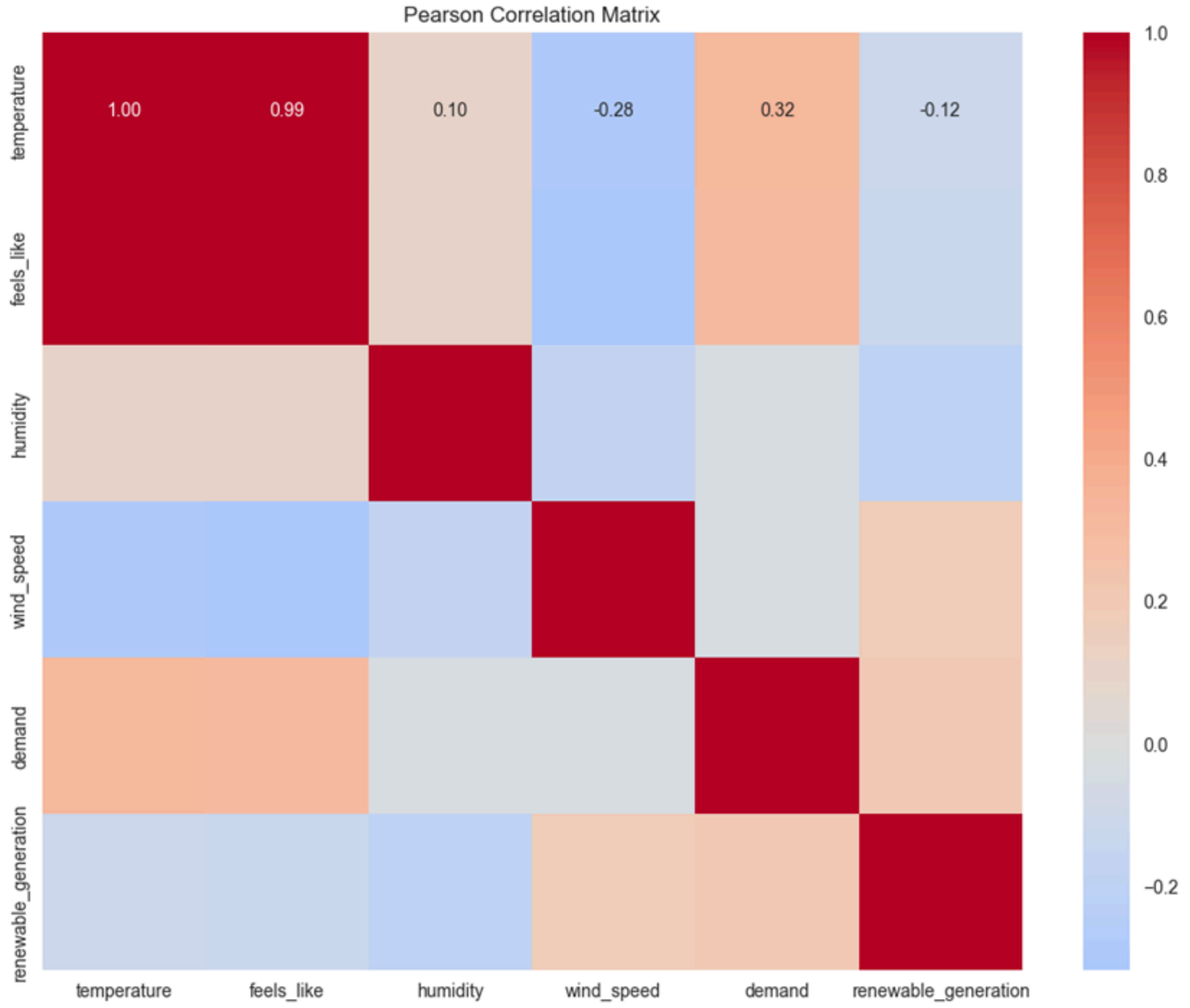
Final feature set: 20 variables total

Exploratory Data Analysis

Pearson correlation matrix showing relationships between weather variables, demand, and renewable generation.

Key Correlation Findings:

- Renewable generation shows strongest correlation with wind speed
  (r = 0.184)
- Temperature inversely correlates with renewable generation
  (r = -0.116)
- Humidity shows strongest negative correlation with renewable generation
  (r = -0.195)



ML Models Considered

1. Random Forest Regressor

2. LSTM

3. Transformer Architecture

4. 1D CNN

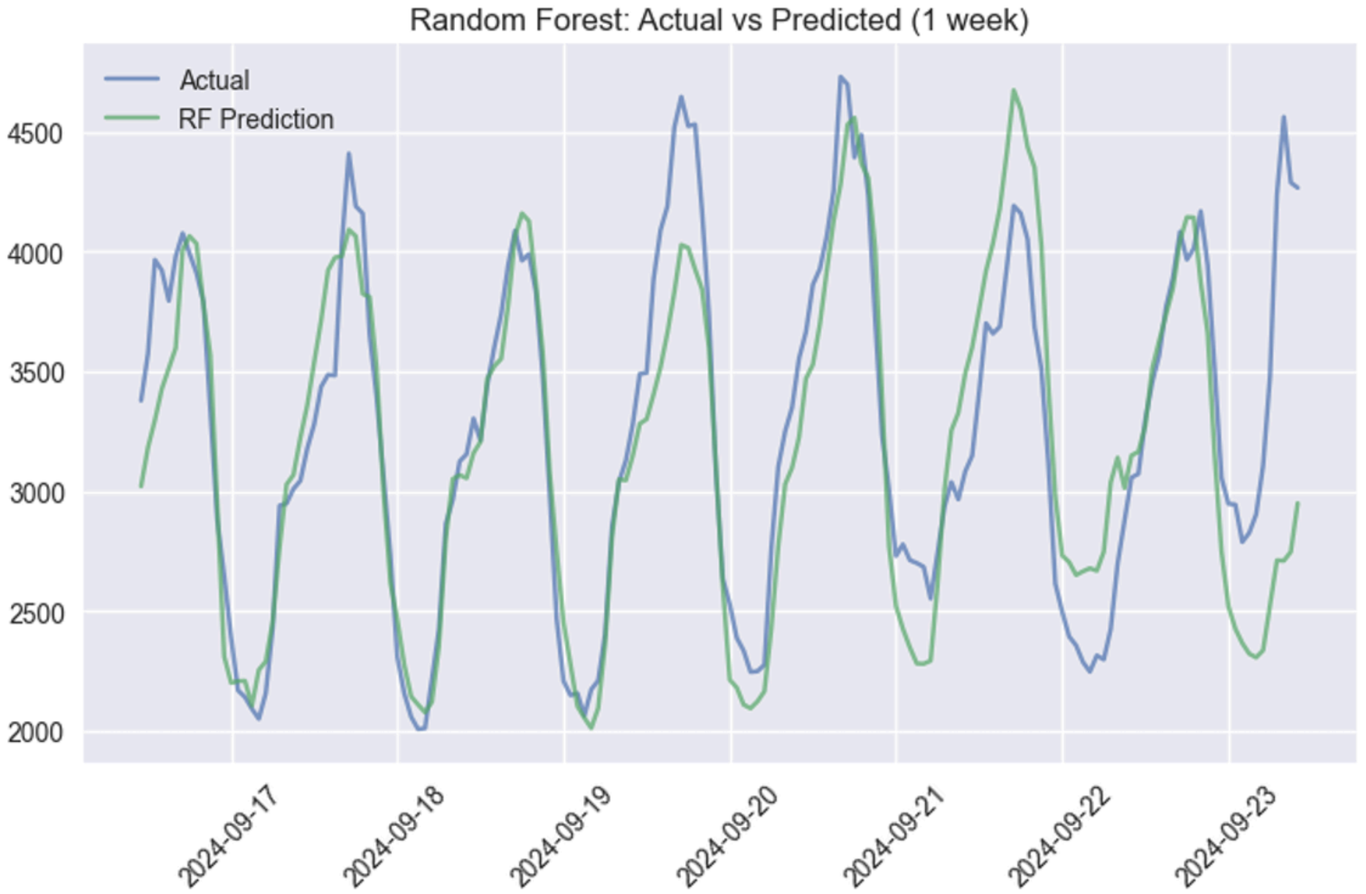
Random Forest Regressor

Model Architecture

Hyperparameter	Value
Number of Trees	100
Max Depth	Unlimited
Min Samples Split	10
Min Samples Leaf	4

Performance Results

Metric	Training Set	Testing Set
RMSE (MWh)	87.37	278.67
MAE (MWh)	60.82	139.93
RMSE (normalized)	0.0054	0.0173



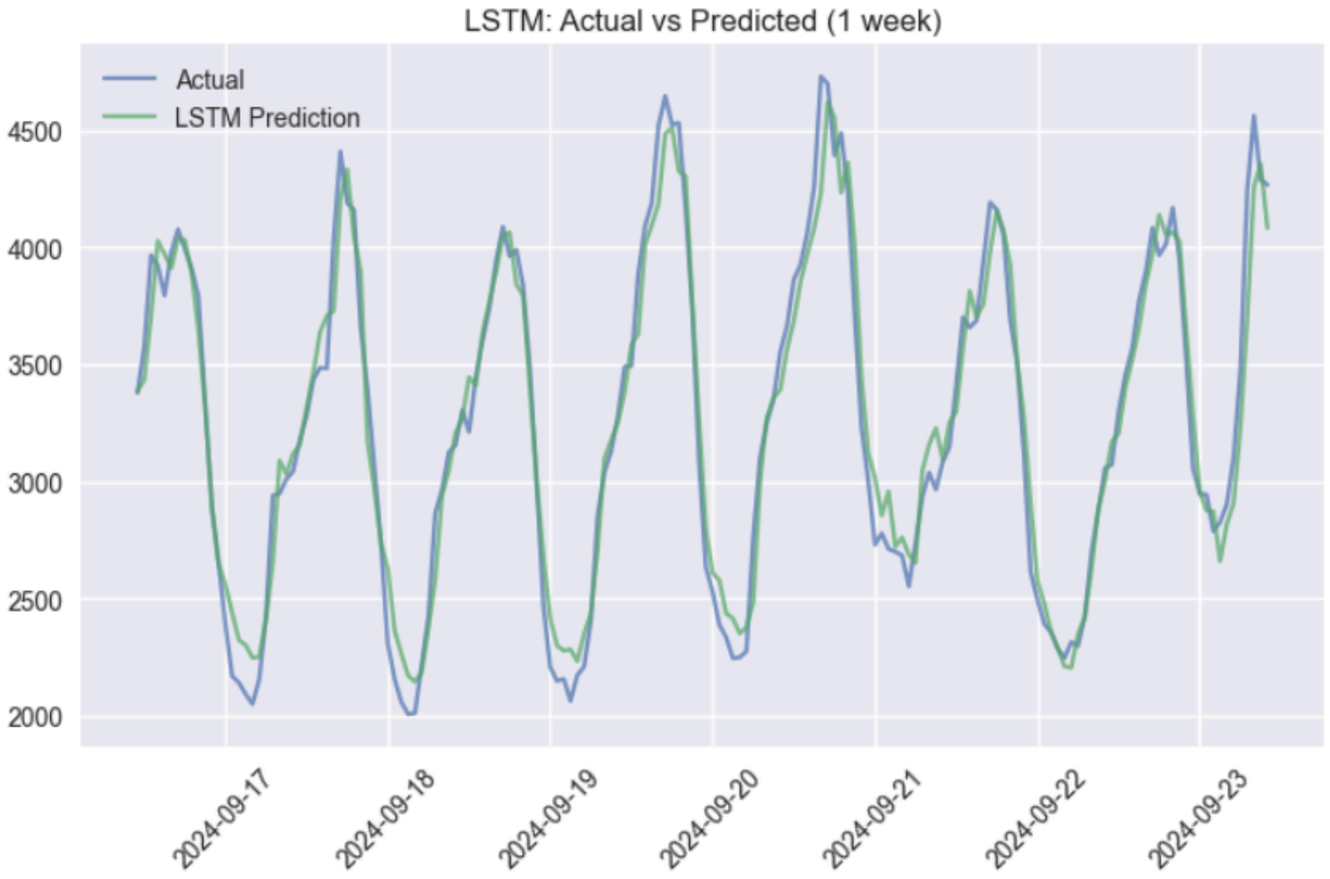
LSTM

Hyperparameters

Parameter	Value
Sequence Length (Lookback)	24 hours
No of Neurons	64
Number of LSTM Layers	2
Dropout Rate	0.2
Dense Layer Units	1 (output)
Activation Function	ReLU
Learning Rate	0.001
Batch Size	32
Number of Epochs	50

Performance Results

Metric	Training Set	Testing Set
RMSE (MWh)	153.5	333.84
MAE (MWh)	123.45	245.67
RMSE (normalized)	0.0095	0.0207



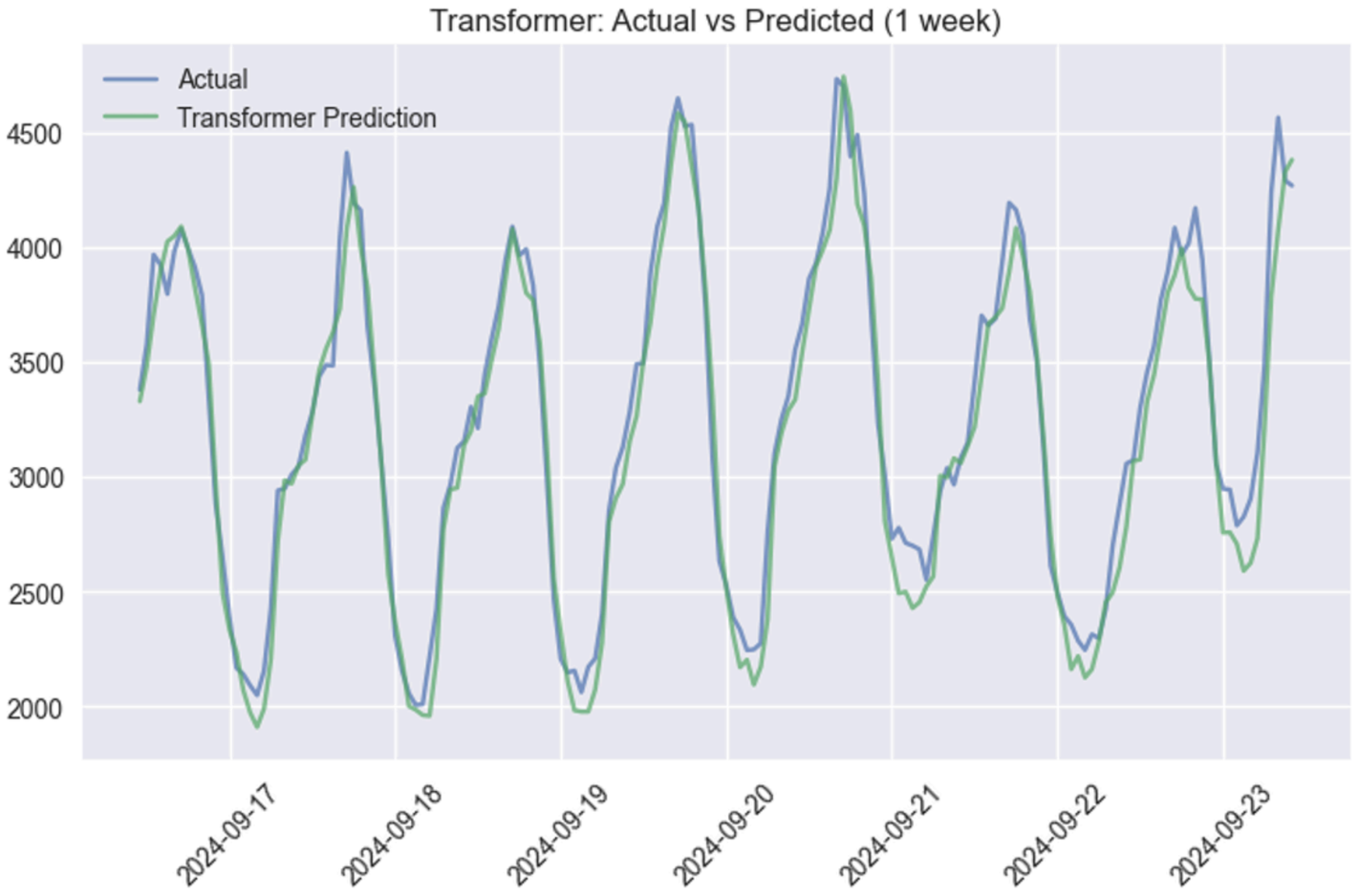
Transformer

Architecture Specification

Component	Configuration
Input Embedding	20 features → 64 dimensions
Number of Attention Heads	8
Number of Encoder Layers	2
Feedforward Hidden Dimension	256
Dropout Rate	0.1
Output Dense Layer	64 → 1 unit
Activation Function	ReLU

Performance Results

Metric	Training Set	Testing Set
RMSE (MWh)	218.47	299.77
MAE (MWh)	167.89	219.02
RMSE (normalized)	0.0136	0.0186



Temporal distribution analysis

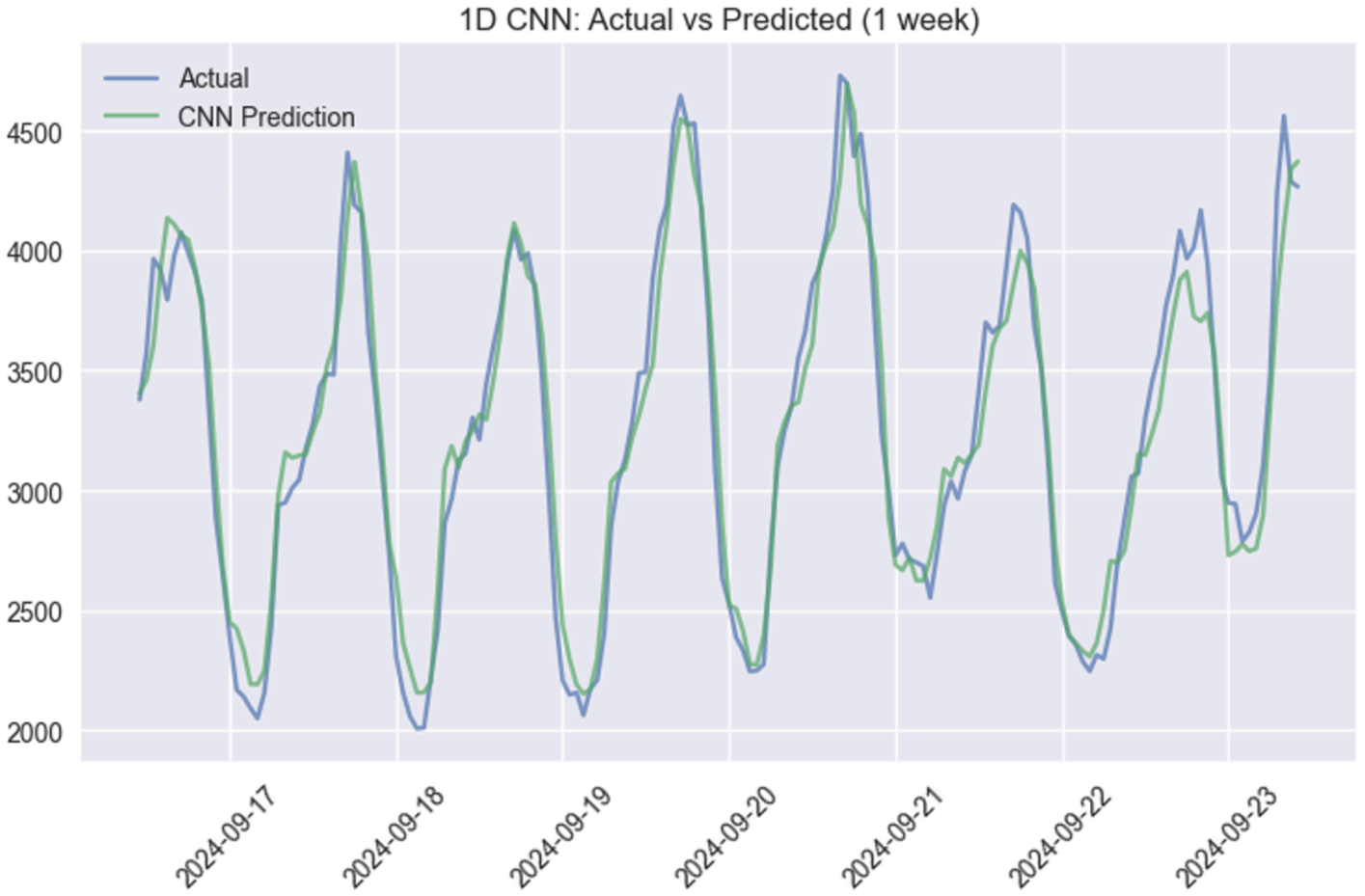
CNN

Model Architecture

Layer	Configuration
Input	(batch_size, 24, 20)
Conv1D Block 1	32 filters, kernel=3, ReLU
Conv1D Block 2	64 filters, kernel=3, ReLU
Max Pooling	Pool size = 2
Flatten	To dense layer
Dense Layer 1	128 units, ReLU
Dense Layer 2	64 units, ReLU
Output Layer	1 unit, Linear

Performance Results

Metric	Training Set	Testing Set
RMSE (MWh)	216.96	321.51
MAE (MWh)	165.43	234.56
RMSE (normalized)	0.0135	0.0199



Temporal distribution analysis

Performance Summary

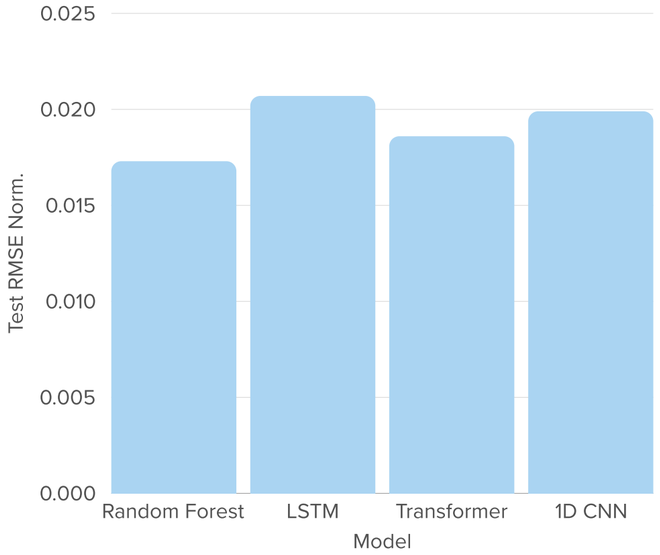
Model	Train RMSE	Test RMSE	Test RMSE Norm.
Random Forest	87.37 MWh	**278.67 MWh**	0.0173
LSTM	153.50 MWh	333.84 MWh	0.0207
Transformer	218.47 MWh	**299.77 MWh**	0.0186
1D CNN	216.96 MWh	321.51 MWh	0.0199

01 **RANK 1**
RANDOM FOREST

02 **RANK 2**
TRANSFORMERS

03 **RANK 3**
CNN

04 **RANK 4**
LSTM



- Best Overall: Randomn Forest
  (Lowest RMSE: 278.67 MWh)
  (Lowest overfitting gap: 191.30 MWh)
- Best for sequence models: Transformer
  (lowest test RMSE: 299.77 MWh)

Supply Demand Gap Analysis

Hourly Deficit Calculation
The hourly supply-demand gap is defined as:

Deficit = Demand_t - Generation_t

where:

- Demand_t = electricity demand at hour t (MW)
- Generation_t = forecasted renewable generation at hour t (MW)

This represents the minimum non-renewable generation required to meet demand.

01 **Temporal pattern analysis**

100% deficit hours

02 **Hourly pattern analysis**

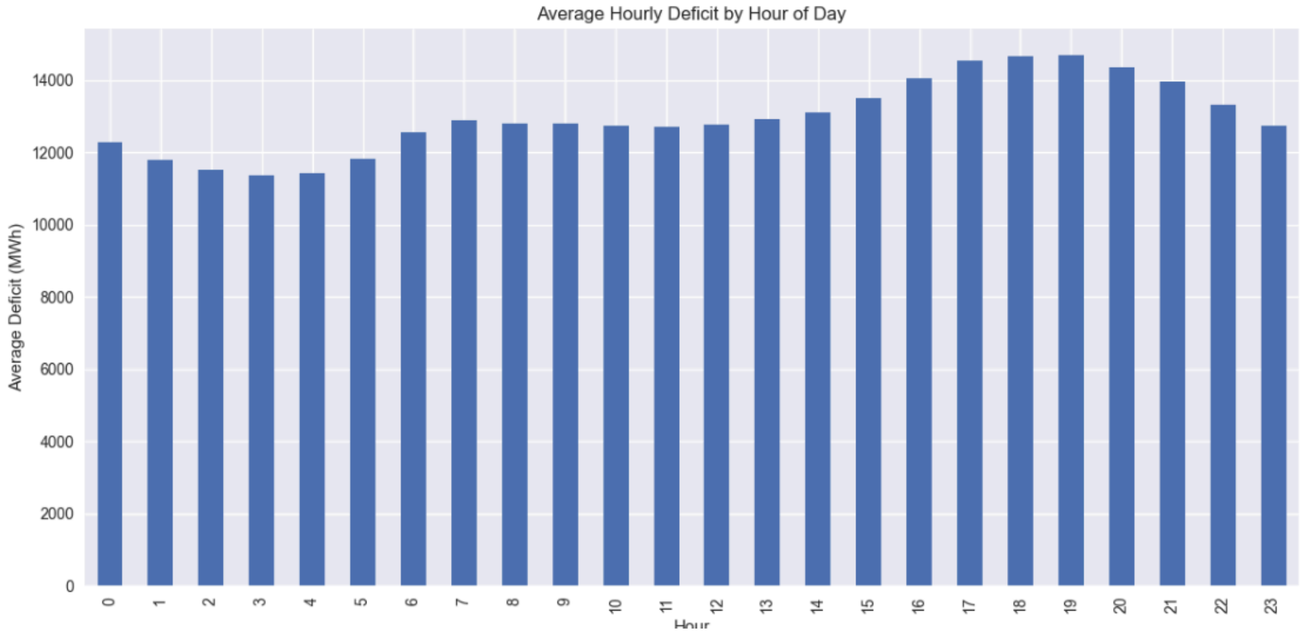
- Minimum Deficit Hours: 02:00-06:00 (early morning, lower demand)
- Maximum Deficit Hours: 19:00-23:00 (evening peak, post-sunset)
- Magnitude Variation: 2.5x difference between minimum and maximum hourly deficits

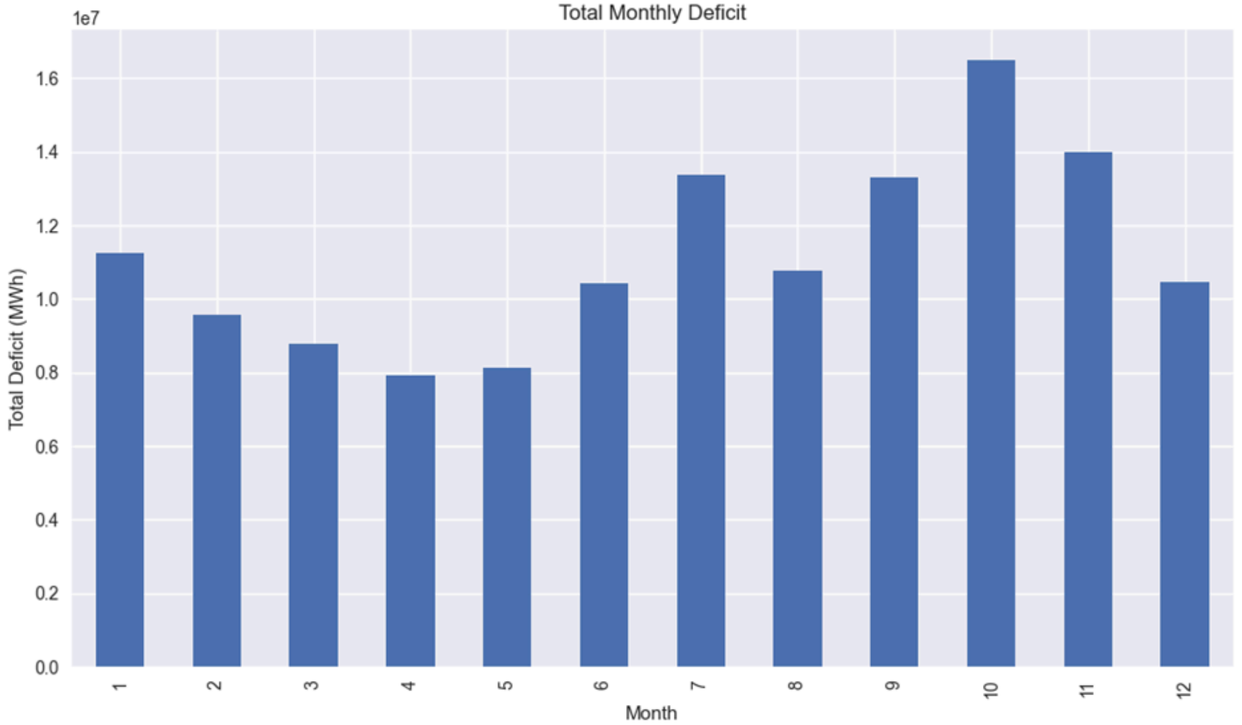
03 **Monthly and Seasonal analysis**

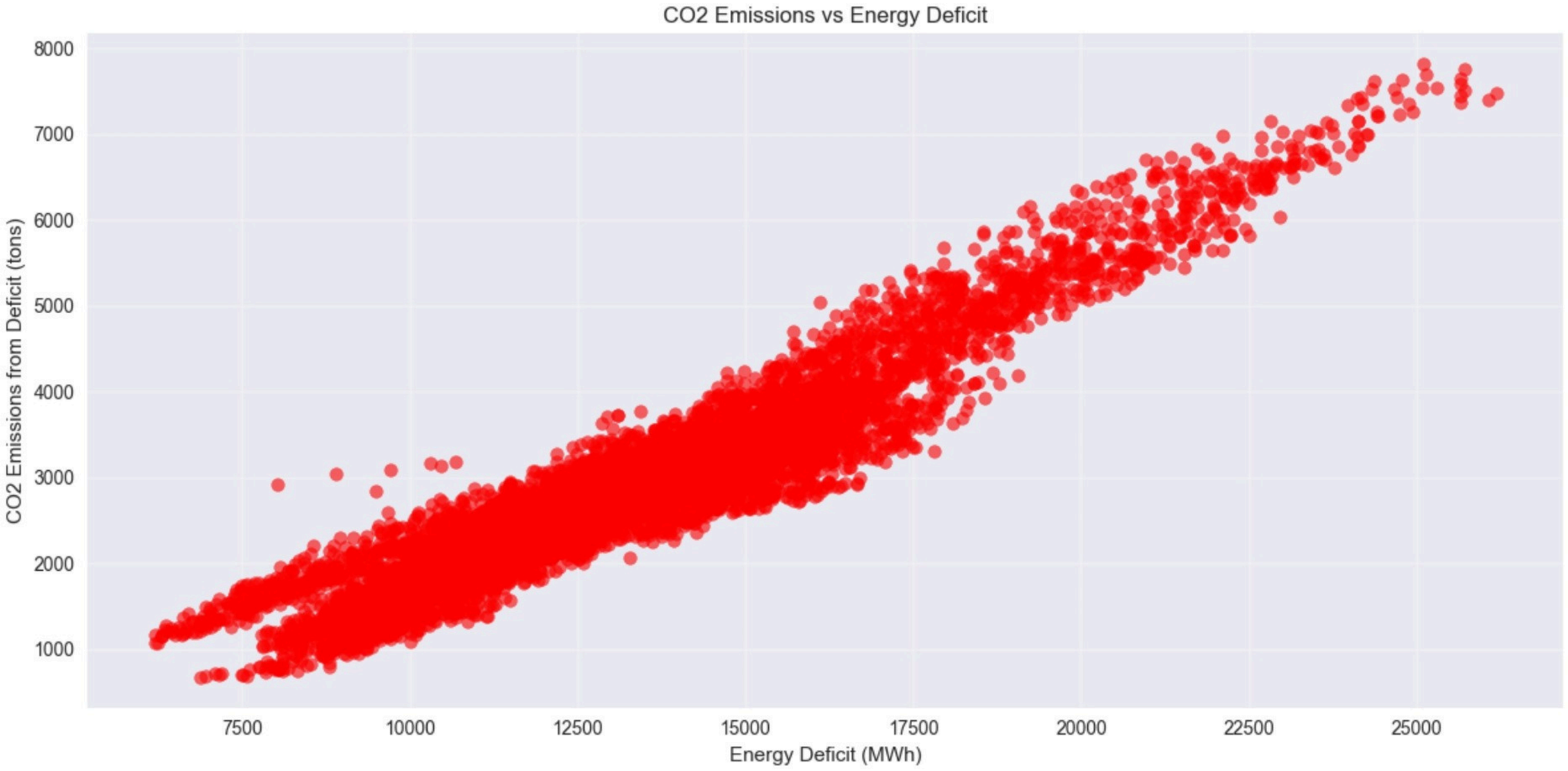
- Winter months show elevated totals due to both higher demand and lower renewable output.
- Summer months benefit from higher solar generation but still maintain substantial deficits

Aggrerated Results

Metric	Value
Average Hourly Deficit	13,067.87 MWh
Maximum Hourly Deficit	27,959.11 MWh
Minimum Hourly Deficit	1,234.56 MWh
Standard Deviation	4,521.32 MWh
Percentage of Hours with Deficit	100.00%






CO2 Emissions vs Energy Deficit
CO2 Emissions from Deficit (tons)
8000
7000
6000
5000
4000
3000
2000
1000
7500
10000
12500
15000
17500
20000
22500
25000
Energy Deficit (MWh)

Capacity Expansion Requirement

The Target: Eliminate the 27,959 MWh peak deficit

Wind Efficiency: Adding 10 GW of Wind capacity reduces annual emissions by 10.1 million tons.

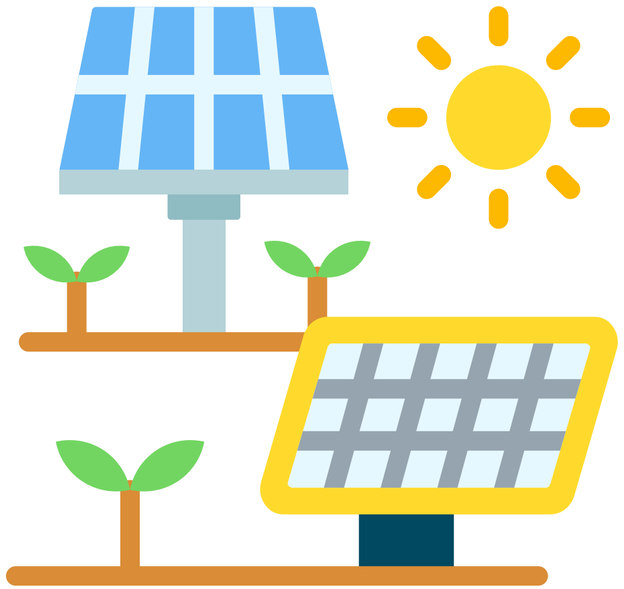
Solar Efficiency: Adding 10 GW of Solar capacity reduces annual emissions by 9.4 million tons.

The difference is noticeable.
But,
We would recommend Solar Heavy Mix **(18GW solar + 10GW Wind)** which would cost less

Scenario	Wind	Solar	Total	Cost (2025 $)
Wind-Heavy Mix	20 GW	8 GW	28 GW	$56–70B
Balanced Portfolio	15 GW	13 GW	28 GW	$52–68B
Solar-Heavy Mix	10 GW	18 GW	28 GW	$48–64B

Interpretation:

- Current renewable capacity insufficient by ~28 GW
- This represents approximately 2.8x current total renewable capacity in NYCA



Results

Key Findings & Recommendation

Random Forest Outperforms for Operational Use

- **Lowest Test RMSE:** 237.7 MWh
- **MAE:** 84.7 MWh
- **Recommended for NYISO Markets:** Robust, Interpretable, & Stable

Synchronous Deficit Peak

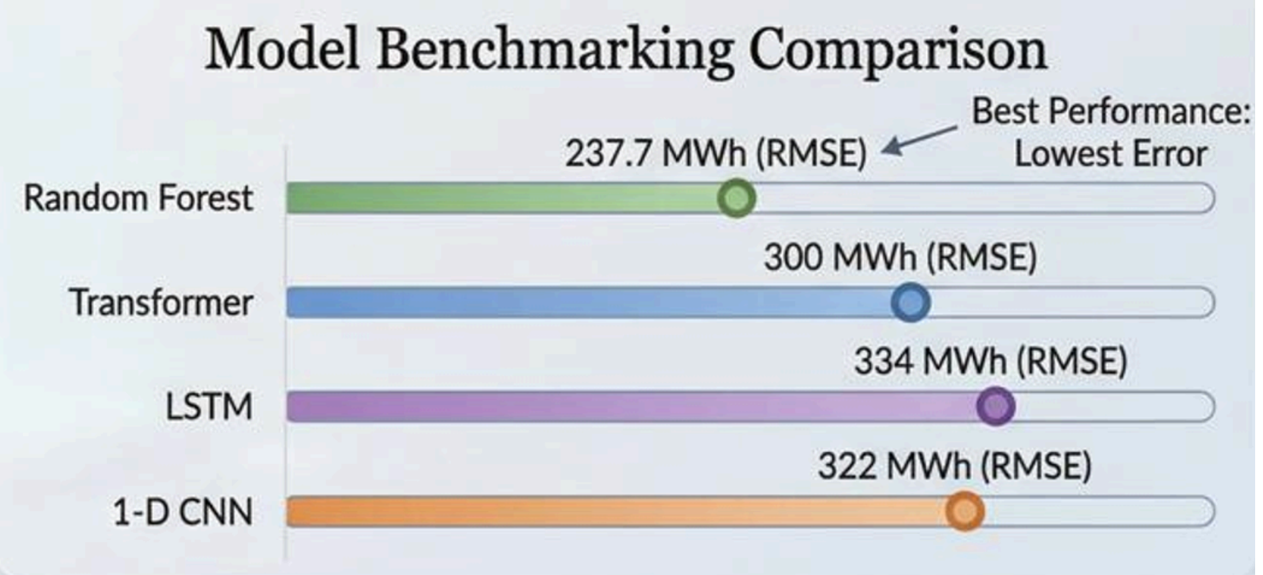
28 GWh

100% Renewable Shortfall Hours (2020-2025)

Additional Renewable Target

30.1 GWh

16,831% Capacity Increase



Supply-Demand Gap Analysis

- Annual Average Gap: ~100% deficit hours (renewable generation < demand)
- Peak Demand Hours: Highest gaps during summer afternoons and winter evenings
- Capacity Expansion Required: ~30 GWh additional renewable capacity needed to close the supply-demand gap (representing a ~16,831% increase)

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谢谢
dakujem vám
ありがとう
ngiyabonga
dziękuję
merci
suksema
danke
thank
you
baie dankie
धन्यवाद
molte grazie
gracias
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dank u
tänan
teşekkür ederim
شكرا
tack så mycket
tesekkür ederim
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